

Automated Data Dictionary System for the MSA ERP Environment

Team Members:

- 1. Shubham Mehta – Project Manager**
- 2. Tanay Gattani – Data Analyst**
- 3. Shubham Jain – Data Engineering**
- 4. Sai Keerthana Jangala – Data Analyst**
- 5. Ajai Shakya – Data Analyst**

Abstract

This project presents a fully automated Data Dictionary system designed for the MSA ERP environment. The system replaces fragmented and manually maintained metadata with a governed, scalable, and continuously updated architecture. Metadata is submitted through a structured Power App interface, validated by administrative reviewers, stored in SQL Server, automatically exported to GitHub, securely transferred via HTTPS, ingested into Microsoft Fabric Lakehouse, curated in Fabric Warehouse, and published as a searchable enterprise Data Dictionary in FreshService. This end-to-end solution improves metadata quality, operational efficiency, and governance readiness while establishing a foundation for long-term enterprise metadata management.

1. Introduction

ERP systems contain large, complex, and interrelated datasets that require clear metadata to support analytics, reporting, compliance, and operational decision-making. Without a unified and governed metadata framework, organizations face ambiguity in interpretation, inconsistencies across business units, and a reliance on undocumented knowledge. Manual spreadsheets, isolated documentation, and inconsistent updates contribute to errors and inefficiencies.

This project addresses these challenges by designing and implementing an automated Data Dictionary system for the MSA ERP ecosystem. The solution integrates user-driven metadata submission, administrative governance, automated ingestion pipelines, secure file transport, curated metadata processing, and enterprise publication. The resulting architecture eliminates manual documentation, increases metadata accuracy, and provides a sustainable mechanism for continuous updates.

2. Background and Related Work

Enterprise metadata management is widely recognized as a foundational requirement for reliable analytics, governance, and regulatory compliance. Data dictionaries and metadata repositories are commonly used to document schema structure, business definitions, ownership, and sensitivity attributes. When maintained effectively, these systems improve data literacy and ensure consistent interpretation of analytical outputs.

Traditional metadata documentation is often managed through spreadsheets or static documents. While easy to create, these approaches do not scale as systems evolve. Schema changes, new data sources, and shifting business definitions frequently result in outdated or inconsistent documentation. Industry frameworks such as the DAMA Data Management Body of Knowledge emphasize that metadata systems must be governed, auditable, and integrated with operational workflows to remain reliable over time.

More recent enterprise solutions include centralized data catalogs that automate schema discovery and lineage tracking. Although technically powerful, these tools often focus on structural metadata and provide limited support for capturing business context directly from subject matter experts. As a result, critical domain knowledge may remain undocumented or disconnected from technical metadata.

This project builds on established metadata management principles by combining user-driven metadata submission with formal governance controls and automated publication. By integrating structured contribution, administrative review, version-controlled storage, and Microsoft Fabric-based ingestion, the system addresses the limitations of both manual documentation and purely automated cataloging approaches while remaining aligned with enterprise metadata best practices. Management practices while addressing the practical limitations observed in the existing MSA ERP documentation process.

3. Project Motivation & Objective

3.1 Project Motivation

Enterprise ERP systems generate large volumes of structured data that are consumed by analysts, engineers, and business stakeholders. In the MSA ERP environment, metadata describing these datasets existed across disconnected spreadsheets and informal documentation maintained by individual teams. This fragmentation made it difficult to identify authoritative definitions, track ownership, or ensure consistent interpretation of fields across reports and dashboards.

A major limitation of the existing approach was the lack of governance. Metadata updates were performed manually without validation, approval workflows, or auditability. As a result, definitions frequently became outdated, schema changes were not reflected in documentation, and

institutional knowledge was lost when personnel changed roles. These issues increased the risk of misinterpretation, reporting errors, and compliance gaps.

The motivation for this project was to replace this ad hoc process with a governed, automated metadata pipeline. By introducing structured submission, administrative validation, version-controlled storage, and automated publication, the system aims to ensure that metadata remains accurate, traceable, and continuously refreshed as the ERP evolves.

3.2 Project Objective

The primary objectives of the Automated Data Dictionary System for the MSA ERP environment are as follows:

1. **Centralized Metadata Contribution**
Design a unified and structured workflow that enables business users to submit metadata through an accessible interface, reducing reliance on fragmented spreadsheets and informal documentation.
2. **Governed Validation and Approval**
Introduce an approval-based governance process in which administrative reviewers validate metadata for completeness, consistency, and alignment with organizational standards before publication.
3. **Authoritative and Auditable Storage**
Establish SQL Server as a secure, version-controlled system of record for approved metadata, supporting traceability, auditability, and historical tracking of changes.
4. **Automated Metadata Generation and Transfer**
Automate the extraction, formatting, and secure transport of approved metadata files to minimize manual intervention and ensure consistent downstream ingestion.
5. **Fabric-Based Ingestion and Curation**
Integrate metadata into Microsoft Fabric Lakehouse and Warehouse layers to support structured ingestion, transformation, and curation within a governed analytical environment.
6. **Enterprise Data Dictionary Accessibility**
Publish a continuously refreshed and searchable Data Dictionary through FreshService, making metadata easily accessible to both technical and non-technical stakeholders.
7. **Scalability and Long-Term Maintainability**
Design the system to be modular, scalable, and maintainable, allowing it to adapt to evolving ERP schemas, governance requirements, and future platform enhancements.

Together, these objectives guide the architectural and implementation decisions of the project and provide clear criteria for evaluating its effectiveness in improving metadata quality, governance, and accessibility.

4. Metadata Sources and Standardization

4.1 Metadata Sources

The system manages business metadata associated with ERP tables and columns. Primary inputs include table names, column names, data types, business definitions, ownership information, and sensitivity classifications. These attributes are provided by business users who possess domain knowledge but may not have technical expertise in database systems.

Additional metadata is generated through the governance process itself, including submission identifiers, approval timestamps, reviewer information, and version indicators. Together, these elements form a complete record of each metadata entry's lifecycle.

4.2 Metadata Validation and Standardization

To ensure consistency, all submissions pass through a structured validation process. Required fields are enforced at the point of submission, preventing incomplete entries. Naming conventions and sensitivity classifications are reviewed by administrators to ensure alignment with organizational standards.

This validation layer serves as a quality gate, ensuring that only approved and standardized metadata progresses downstream. By separating submission from approval, the system maintains data quality while preserving accessibility for contributors.

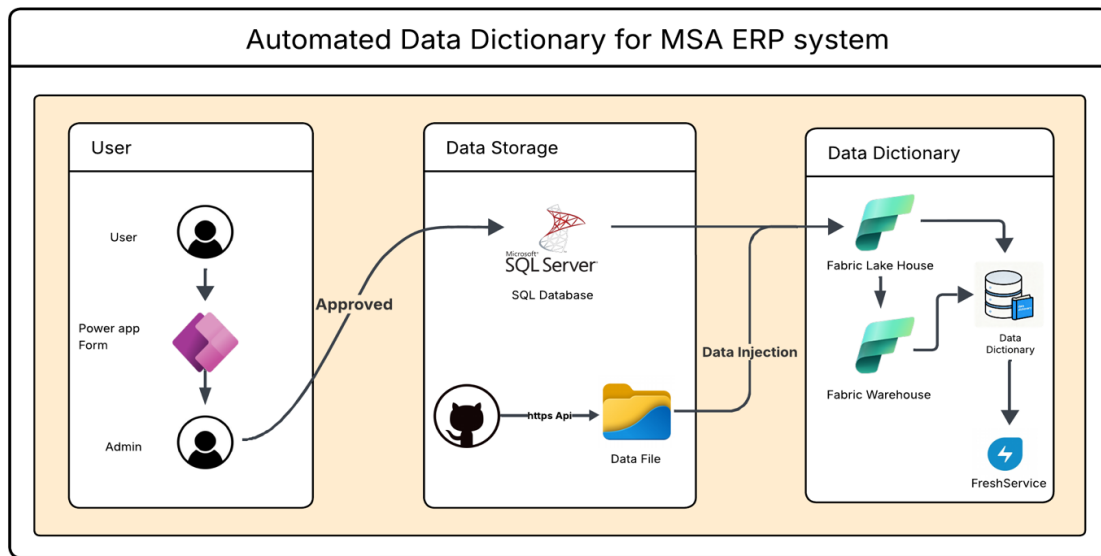
The screenshot displays a Power Apps interface titled "WISCONSIN UNIVERSITY OF MADISON" and "Power Apps | UserInterface (Editing)". A green notification bar at the top states "Feedback submitted for review." The main form area is titled "Filter by Table Name : vw_ActualData" and includes a "Search by Column Name" input field. The form contains a list of columns on the left, each with a right-pointing chevron: "ActualInvoiceAmount_vw_ActualData", "ActualHours_vw_ActualData", "ActualAmount_vw_ActualData", "ActualAmount_FeeMgt_vw_ActualData", "ActualAmount_Expense_vw_ActualData", and "ActualAmount_Labor_vw_ActualData". To the right of this list, there are three text input fields: "Description" (containing "This column stores the monetary amount related to actualinvoiceamount."), "Feedback" (containing "This is a Test Description"), and "Notes". A blue "Save" button is located at the bottom right of the form.

4.2.1 Power Apps Submission Form

5. System Architecture and Design Decisions

5.1 Overall Architecture

The system follows a multi-layer architecture that models the complete lifecycle of metadata from submission to publication. Each layer has a clearly defined responsibility, allowing the system to scale and evolve without tightly coupled dependencies. The architecture intentionally separates operational workflows from analytical processing to maintain governance boundaries.

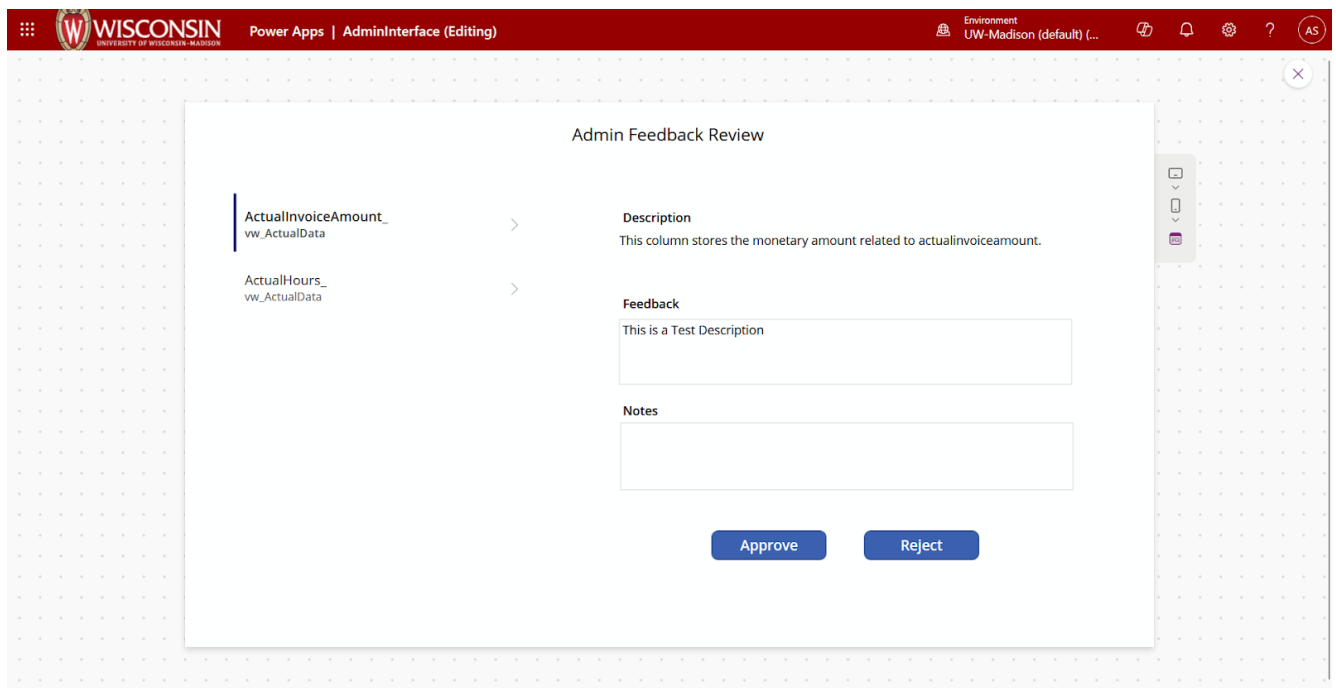


5.1.1 Architecture Diagram

5.2 Submission and Governance Layer

Metadata submission is handled through a Power Apps interface designed for non-technical users. The interface enforces required fields and standardized formats while minimizing cognitive overhead. Once submitted, entries are automatically marked as pending and routed to administrators for review.

Administrative review acts as the central governance mechanism. Reviewers assess completeness, naming consistency, definition clarity, and sensitivity classification. Approved entries proceed to storage, while rejected entries are returned with feedback. This step ensures accountability and prevents unvalidated metadata from entering the system.



5.2.1 Power Apps Admin Interface

5.2.2 System State Before Administrative Approval and After User Feedback

ID	TableName	ColumnName	Description	Feedback	Notes	FeedbackStatus
1	vw_ActualData	ActualInvoiceAmount	This column stores the monetary amount related to...	This is a Test Description	This is a Test Case	Pending
2	vw_ActualData	ActualHours	This column stores the number of hours related to...			Pending
3	vw_ActualData	ActualAmount	This column stores the monetary amount related to...			NULL
4	vw_ActualData	ActualAmount_FeeMgt	This column stores the monetary amount related to...			NULL
5	vw_ActualData	ActualAmount_Expense	This column stores the monetary amount related to...	NULL	NULL	NULL
6	vw_ActualData	ActualAmount_Labor	This column stores the monetary amount related to...	NULL	NULL	NULL
7	vw_ActualData	ActualAmount_Revenue	This column stores the monetary amount related to...	NULL	NULL	NULL
8	vw_ActualData	ActualAmount_WWP	This column stores the monetary amount related to...	NULL	NULL	NULL
9	vw_ActualData	CategoryID	This column stores the unique identifier for category...	NULL	NULL	NULL
10	vw_ActualData	FKDateFinancial	This column records the date associated with foreign...	NULL	NULL	NULL
11	vw_ActualData	FKDateSales	This column stores information about foreign key...	NULL	NULL	NULL
12	vw_ActualData	FKProject	This column relates to project details for foreign k...	NULL	NULL	NULL
13	vw_ActualData	FKCustomer	This column references the customer information...	NULL	NULL	NULL
14	vw_ActualData	FKProjectManager	This column identifies the manager responsible fo...	NULL	NULL	NULL
15	vw_ActualData	FKProjTransDate	This column records the date associated with forei...	NULL	NULL	NULL
16	vw_ActualData	FKResource	This column identifies the resource or person ass...	NULL	NULL	NULL
17	vw_ActualData	FKSalesManager	This column identifies the manager responsible fo...	NULL	NULL	NULL
18	vw_ActualData	FKSalesCurrency	This column specifies the currency used for foreig...	NULL	NULL	NULL
19	vw_ActualData	FKTimeSheet	This column stores information about foreign key...	NULL	NULL	NULL
20	vw_ActualData	LedgerTransDate	This column records the date associated with ledg...	NULL	NULL	NULL
21	vw_ActualData	FKFinancialDimension	This column stores information about foreign key...	NULL	NULL	NULL
22	vw_ActualData	FKMainAccount	This column stores information about foreign key...	NULL	NULL	NULL
23	vw_ActualData	ActualHours_Total	This column stores the number of hours related to...	NULL	NULL	NULL
24	vw_ActualSal	PKActualSales	This column stores information about pkActualSales...	NULL	NULL	NULL
25	vw_ActualSal	FKSales	This column stores information about foreign key...	NULL	NULL	NULL
26	vw_ActualSal	FKSales	This column stores information about foreign key...	NULL	NULL	NULL
27	vw_ActualSal	FKProject	This column relates to project details for foreign k...	NULL	NULL	NULL
28	vw_ActualSal	FKDateFinancial	This column records the date associated with forei...	NULL	NULL	NULL
29	vw_ActualSal	FKProjectFundingSource	This column relates to project details for foreign k...	NULL	NULL	NULL
30	vw_ActualSal	FKCustomer	This column references the customer information...	NULL	NULL	NULL
31	vw_ActualSal	FKSalesCurrency	This column specifies the currency used for foreig...	NULL	NULL	NULL
32	vw_ActualSal	FKProjectCategory	This column relates to project details for foreign k...	NULL	NULL	NULL
33	vw_ActualSal	FKProjectManager	This column identifies the manager responsible fo...	NULL	NULL	NULL
34	vw_ActualSal	FKSalesManager	This column identifies the manager responsible fo...	NULL	NULL	NULL
35	vw_ActualSal	FKResource	This column identifies the resource or person ass...	NULL	NULL	NULL
36	vw_ActualSal	FKProjTransDate	This column records the date associated with forei...	NULL	NULL	NULL
37	vw_ActualSal	FKWorkLocation	This column stores information about foreign key...	NULL	NULL	NULL
38	vw_ActualSal	FKProjectController	This column relates to project details for foreign k...	NULL	NULL	NULL

ID	TableName	ColumnName	Description	Feedback	Notes	FeedbackStatus
1	vw_ActualData	ActualInvoiceAmount	This column stores the monetary amount related to actualinvoiceamount...	This is a Test Description	This is a Test Case	Pending
2	vw_ActualData	ActualHours	This column stores the number of hours related to actualhours...			Pending
3	vw_ActualData	ActualAmount	This column stores the monetary amount related to actualamount...			NULL
4	vw_ActualData	ActualAmount_FeeMgt	This column stores the monetary amount related to actualamount fe...			NULL
5	vw_ActualData	ActualAmount_Expense	This column stores the monetary amount related to actualamount ex...			NULL
6	vw_ActualData	ActualAmount_Labor	This column stores the monetary amount related to actualamount la...			NULL
7	vw_ActualData	ActualAmount_Revenue	This column stores the monetary amount related to actualamount re...			NULL
8	vw_ActualData	ActualAmount_WWP	This column stores the monetary amount related to actualamount w...			NULL
9	vw_ActualData	CategoryID	This column stores the unique identifier for categoryid...			NULL
10	vw_ActualData	FKDateFinancial	This column records the date associated with foreign key datefinan...			NULL
11	vw_ActualData	FKDateSales	This column stores information about foreign key datesales...			NULL
12	vw_ActualData	FKProject	This column relates to project details for foreign key project...			NULL
13	vw_ActualData	FKCustomer	This column references the customer information for foreign key cus...			NULL
14	vw_ActualData	FKProjectManager	This column identifies the manager responsible for foreign key proje...			NULL
15	vw_ActualData	FKProjTransDate	This column records the date associated with foreign key project...			NULL
16	vw_ActualData	FKResource	This column identifies the resource or person associated with foreig...			NULL
17	vw_ActualData	FKSalesManager	This column identifies the manager responsible for foreign key sales...			NULL
18	vw_ActualData	FKSalesCurrency	This column specifies the currency used for foreign key salescurrency...			NULL
19	vw_ActualData	FKTimeSheet	This column stores information about foreign key timesheet...			NULL
20	vw_ActualData	LedgerTransDate	This column records the date associated with ledgertransdate...			NULL
21	vw_ActualData	FKFinancialDimension	This column stores information about foreign key financialdimension...			NULL
22	vw_ActualData	FKMainAccount	This column stores information about foreign key mainaccount...			NULL
23	vw_ActualData	ActualHours_Total	This column stores the number of hours related to actualhours total...			NULL

5.2.2.1 SQL Database Feedback Status Changes to "Pending"

5.2.2.2 Fabric Data Flow Feedback Status Changes to "Pending"

5.3 SQL Server as the Operational Metadata Store

SQL Server serves as the system of record for all approved metadata. Each entry includes governance attributes such as approval timestamps, reviewer identity, and version information. This structured storage enables auditability and reliable extraction for automation processes.

By isolating operational metadata storage from analytical platforms, the system ensures that governance workflows do not interfere with downstream processing.

5.3.1 System State After Administrative Approval

ID	TableName	ColumnName	Description	Feedback	Notes	FeedbackStatus
1	vw_ActualData	ActualInvoiceAmount	This is a Test Description	NULL		Approved
2	vw_ActualData	ActualHours	This is a Test Case	NULL		Approved
3	vw_ActualData	ActualAmount	This column stores the monetary amount related to actualamount.	NULL		NULL
4	vw_ActualData	ActualAmount_FeeMgt	This column stores the monetary amount related to actualamount fee mgt.	NULL		NULL
5	vw_ActualData	ActualAmount_Expense	This column stores the monetary amount related to actualamount expense.	NULL	NULL	NULL
6	vw_ActualData	ActualAmount_Labor	This column stores the monetary amount related to actualamount labor.	NULL	NULL	NULL
7	vw_ActualData	ActualAmount_Revenue	This column stores the monetary amount related to actualamount revenue.	NULL	NULL	NULL
8	vw_ActualData	ActualAmount_WIP	This column stores the monetary amount related to actualamount wip.	NULL	NULL	NULL
9	vw_ActualData	CategoryID	This column stores the unique identifier for category.	NULL	NULL	NULL
10	vw_ActualData	FCDateFinancial	This column records the date associated with foreign key datefinancial.	NULL	NULL	NULL
11	vw_ActualData	FCDataArea	This column stores information about foreign key dataarea.	NULL	NULL	NULL
12	vw_ActualData	FCProject	This column relates to project details for foreign key project.	NULL	NULL	NULL
13	vw_ActualData	FCCustomer	This column references the customer information for foreign key customer.	NULL	NULL	NULL
14	vw_ActualData	FCProjectManager	This column identifies the manager responsible for foreign key projectmanager.	NULL	NULL	NULL
15	vw_ActualData	FCProjTransDate	This column records the date associated with foreign key projtransdate.	NULL	NULL	NULL
16	vw_ActualData	FCResource	This column identifies the resource or person associated with foreign key resource.	NULL	NULL	NULL
17	vw_ActualData	FCSalesManager	This column identifies the manager responsible for foreign key salesmanager.	NULL	NULL	NULL
18	vw_ActualData	FCSalesCurrency	This column specifies the currency used for foreign key salescurrency.	NULL	NULL	NULL
19	vw_ActualData	FCTimeSheet	This column stores information about foreign key timesheet.	NULL	NULL	NULL
20	vw_ActualData	LedgerTransDate	This column records the date associated with ledgertransdate.	NULL	NULL	NULL
21	vw_ActualData	FCFinancialDimension	This column stores information about foreign key financialdimension.	NULL	NULL	NULL
22	vw_ActualData	FCMainAccount	This column stores information about foreign key mainaccount.	NULL	NULL	NULL
23	vw_ActualData	ActualHours_Total	This column stores the number of hours related to actualhours total.	NULL	NULL	NULL
24	vw_ActualData	FCSalesActualData	This column stores information about salesactualdata.	NULL	NULL	NULL
25	vw_ActualData	FCActivity	This column stores information about foreign key activity.	NULL	NULL	NULL
26	vw_ActualData	FCDataArea	This column stores information about foreign key dataarea.	NULL	NULL	NULL
27	vw_ActualData	FCProject	This column relates to project details for foreign key project.	NULL	NULL	NULL
28	vw_ActualData	FCDateFinancial	This column records the date associated with foreign key datefinancial.	NULL	NULL	NULL
29	vw_ActualData	FCProjectFundingSource	This column relates to project details for foreign key projectfundingsource.	NULL	NULL	NULL
30	vw_ActualData	FCCustomer	This column references the customer information for foreign key customer.	NULL	NULL	NULL
31	vw_ActualData	FCSalesCurrency	This column specifies the currency used for foreign key salescurrency.	NULL	NULL	NULL
32	vw_ActualData	FCProjectCategory	This column relates to project details for foreign key projectcategory.	NULL	NULL	NULL
33	vw_ActualData	FCProjectManager	This column identifies the manager responsible for foreign key projectmanager.	NULL	NULL	NULL

ID	TableName	ColumnName	Description	Feedback	Notes	FeedbackStatus
1	vw_ActualData	ActualInvoiceAmount	This is a Test Description	NULL		Approved
2	vw_ActualData	ActualHours	This is a Test Case	NULL		Approved
3	vw_ActualData	ActualAmount	This column stores the monetary amount related to actualamount.	NULL		NULL
4	vw_ActualData	ActualAmount_FeeMgt	This column stores the monetary amount related to actualamount.	NULL		NULL
5	vw_ActualData	ActualAmount_Expense	This column stores the monetary amount related to actualamount.	NULL	NULL	NULL
6	vw_ActualData	ActualAmount_Labor	This column stores the monetary amount related to actualamount.	NULL	NULL	NULL
7	vw_ActualData	ActualAmount_Revenue	This column stores the monetary amount related to actualamount.	NULL	NULL	NULL
8	vw_ActualData	ActualAmount_WIP	This column stores the monetary amount related to actualamount.	NULL	NULL	NULL
9	vw_ActualData	CategoryID	This column stores the unique identifier for category.	NULL	NULL	NULL
10	vw_ActualData	FCDateFinancial	This column records the date associated with foreign key datefinancial.	NULL	NULL	NULL
11	vw_ActualData	FCDataArea	This column stores information about foreign key dataarea.	NULL	NULL	NULL
12	vw_ActualData	FCProject	This column relates to project details for foreign key project.	NULL	NULL	NULL
13	vw_ActualData	FCCustomer	This column references the customer information for foreign key customer.	NULL	NULL	NULL
14	vw_ActualData	FCProjectManager	This column identifies the manager responsible for foreign key projectmanager.	NULL	NULL	NULL
15	vw_ActualData	FCProjTransDate	This column records the date associated with foreign key projtransdate.	NULL	NULL	NULL
16	vw_ActualData	FCResource	This column identifies the resource or person associated with foreign key resource.	NULL	NULL	NULL
17	vw_ActualData	FCSalesManager	This column identifies the manager responsible for foreign key salesmanager.	NULL	NULL	NULL
18	vw_ActualData	FCSalesCurrency	This column specifies the currency used for foreign key salescurrency.	NULL	NULL	NULL
19	vw_ActualData	FCTimeSheet	This column stores information about foreign key timesheet.	NULL	NULL	NULL
20	vw_ActualData	LedgerTransDate	This column records the date associated with ledgertransdate.	NULL	NULL	NULL
21	vw_ActualData	FCFinancialDimension	This column stores information about foreign key financialdimension.	NULL	NULL	NULL
22	vw_ActualData	FCMainAccount	This column stores information about foreign key mainaccount.	NULL	NULL	NULL
23	vw_ActualData	ActualHours_Total	This column stores the number of hours related to actualhours total.	NULL	NULL	NULL

5.3.1.1 SQL Database once the “Feedback” is approved

5.3.1.2 Fabric once the “Feedback” is approved

5.4 Secure Transfer and Ingestion

Metadata files are transferred from GitHub to Microsoft Fabric using an HTTPS-based API. This transfer layer ensures encrypted transmission, authenticated access, and logging of success and failure events. Secure transport is critical to maintaining data integrity and compliance.

6. Metadata Processing in Microsoft Fabric

6.1 Lakehouse Ingestion Layer

Within Microsoft Fabric, metadata files are first landed in the Lakehouse as raw assets. Schema validation and parsing occur at this stage to ensure structural integrity. The Lakehouse functions as a staging zone where metadata is standardized prior to transformation.

```

1 import os
2 from functools import reduce
3 from pyspark.sql import SparkSession, DataFrame
4 from pyspark.sql.functions import lit, current_timestamp

[1] ✓ - Session ready in 12 sec 995 ms. Command executed in 531 ms by TANAY GATTANI on 11:48:55 PM, 11/30/25

...

1 path = 'Files/RawDataMSA_Data'
2 description_source_table = "dictionary"
3 dictionary_table_name = "data_dictionary"
4 all_data_dicts = []
5
6 try:
7     files = mssparkutils.fs.ls(path)
8
9     for file_info in files:
10         if file_info.name.endswith('.csv'):
11
12             file_path = file_info.path
13             table_name = file_info.name.replace('.csv', '')
14
15             df = spark.read.csv(file_path, header=True, inferSchema=True)
16
17             # Register as temp SQL
18             temp_view_name = "temp_source_data"
19             df.createOrReplaceTempView(temp_view_name)
20
21             schema_df = spark.sql(f"DESCRIBE {temp_view_name}")
22
23             columns_to_profile = schema_df.collect()

```

6.1.1 script for data dictionary.

6.2 Warehouse Curation Layer

After ingestion, metadata is transformed into curated Warehouse tables. This process includes standardizing data types, integrating governance attributes, and applying versioning logic. Reference tables and dictionary views are constructed to support efficient querying and publication.

The Warehouse serves as the authoritative source for enterprise metadata, supporting both analytical consumption and external publication.

Queries [43]

- vw_ActualData csv
- vw_ActualSales csv
- vw_AnnualDirectWa...
- vw_AnnualRevenue ...
- vw_Budget csv
- vw_BudgetSales csv
- vw_CEODashboardD...
- vw_Cashreceipts csv
- vw_ClientData csv
- vw_ClientLiasion csv
- vw_ContractAgree...
- vw_Customer csv
- vw_CustomerTransa...
- vw_CustomerTransa...

Query settings

Properties

Name: vw_ActualData csv

Applied steps

- Source
- Navigation
- Navigation 1
- Navigation 2
- Navigation 3
- Navigation 4
- Imported C...

Data destination

No data destination

Completed (1.40 s) Columns: 23 Rows: 6 Add default destination... Not run

6.2 Warehouse

7. Data Dictionary Publication and Access

The finalized Data Dictionary is published through FreshService, providing a searchable interface for organizational users. Stakeholders can search by table, column, or business domain and view detailed definitions, ownership information, sensitivity classifications, and update timestamps.

This publication layer ensures that metadata is accessible and meaningful to both technical and non-technical users. By integrating with an existing enterprise platform, the system avoids introducing additional tools while maximizing adoption.

data_dictionary

Showing 450 rows

Table view

Table_name	Column_name	Data_Type	Description	Null_Count	Not_Null_Cou...	Total_Rows	Null_Percenta...	Unique_Values	Sample_Value	Generated_D...
vw_ActualData	ActualAmount_	double	This column store...	0	5	5	0	5	-973.9	2025-12-05T17:01.
vw_ActualData	FKDataArea	string	This column store...	0	5	5	0	1	egt	2025-12-05T17:01.
vw_ActualData	CategoryId	string	This column store...	0	5	5	0	1	Eyzas	2025-12-05T17:01.
vw_ActualData	ActualInvoiceAmo...	double	This is a test case	0	5	5	0	5	957.2	2025-12-05T17:01.
vw_ActualData	FKDateFinancial	string	This column recor...	0	5	5	0	1	fxp92071032	2025-12-05T17:01.
vw_ActualData	ActualAmount_WIP_	double	This column store...	0	5	5	0	1	8.7	2025-12-05T17:01.
vw_ActualData	ActualAmount_Ex...	double	This column store...	0	5	5	0	1	5.4	2025-12-05T17:01.
vw_ActualData	FKProjTransDate	int	This column recor...	0	5	5	0	5	77575864	2025-12-05T17:01.
vw_ActualData	FKProject	string	This column relate...	0	5	5	0	4	vflq568628173.09...	2025-12-05T17:01.
vw_ActualData	FKResource	bigint	This column identi...	0	5	5	0	4	9578452075	2025-12-05T17:01.
vw_ActualData	ActualAmount_Fe...	double	This is a test case	0	5	5	0	1	9.2	2025-12-05T17:01.
vw_ActualData	ActualAmount_La...	double	This column store...	0	5	5	0	1	4.9	2025-12-05T17:01.
vw_ActualData	FKCustomer	string	This column refere...	0	5	5	0	1	Parker Carter	2025-12-05T17:01.
vw_ActualData	FKSalesManager	bigint	This column identi...	0	5	5	0	1	3785940984	2025-12-05T17:01.
vw_ActualData	ActualHours_	double	This column store...	0	5	5	0	1	2.3	2025-12-05T17:01.
vw_ActualData	ActualAmount_Re...	double	This column store...	0	5	5	0	5	755.5	2025-12-05T17:01.
vw_ActualData	FKSalesCurrency	string	This column specif...	0	5	5	0	1	RTS	2025-12-05T17:01.
vw_ActualData	ActualHours_Total_	double	This column store...	0	5	5	0	1	9.8	2025-12-05T17:01.
vw_ActualData	FKMainAccount	bigint	This column store...	0	5	5	0	1	4267744751	2025-12-05T17:01.
vw_ActualData	FKFinancialDimen...	string	This column store...	0	5	5	0	4	YMF0432366287	2025-12-05T17:01.
vw_ActualData	LedgerTransDate	date	This column recor...	0	5	5	0	1	6153-03-03	2025-12-05T17:01.

Succeeded (17 sec 447 ms)

Columns 11 Rows 450

7.1 Data Dictionary

8. Evaluation and System Effectiveness

8.1 Evaluation Criteria

The Automated Data Dictionary System was evaluated based on four primary criteria: metadata accuracy, governance reliability, scalability, and operational efficiency. Rather than relying on formal user studies or quantitative benchmarks, the evaluation focused on whether the system effectively fulfills its stated project objectives and supports reliable, governed metadata management within the MSA ERP environment.

These criteria were selected to emphasize practical effectiveness, maintainability, and enterprise applicability, rather than system performance metrics such as throughput or latency.

8.2 Governance and Metadata Accuracy

From a governance perspective, the introduction of an approval-based validation workflow significantly improves metadata accuracy. Administrative review ensures that published definitions reflect business-approved interpretations rather than informal or inconsistent descriptions. This reduces ambiguity and increases trust in analytical outputs derived from ERP data.

The inclusion of governance attributes such as approval timestamps, reviewer identity, and version information further strengthens accountability. Together, these mechanisms ensure that metadata remains reliable, auditable, and aligned with organizational standards.

8.3 Scalability and Operational Efficiency

The system's modular architecture supports scalability by allowing new tables, columns, and metadata domains to be added without requiring architectural changes. Each layer in the pipeline has a clearly defined responsibility, enabling the system to grow alongside the ERP environment. Automation plays a critical role in improving operational efficiency. By replacing manual documentation updates with automated extraction, transfer, and ingestion processes, the system minimizes repetitive effort and reduces the risk of outdated documentation. As a result, metadata remains synchronized with evolving schemas while reducing reliance on subject matter experts for routine maintenance.

8.4 Practical Applicability and Overall Assessment

Overall, the system successfully meets its intended objectives by delivering a governed, automated, and enterprise-ready Data Dictionary solution. While it is not designed to replace advanced data governance platforms or lineage tools, it fills a critical gap by providing a reliable mechanism for capturing, validating, and publishing business metadata.

The system demonstrates that combining user-driven contribution with formal governance controls and automated publication can significantly improve metadata quality and accessibility. This makes the solution both practically valuable and extensible for future governance and analytics initiatives.

9. Limitations

The system's effectiveness is heavily dependent on the quality and consistency of user submissions as well as the thoroughness of administrative review. Incomplete, inaccurate, or inconsistent submissions can impact overall reliability and the usefulness of the information presented. External dependencies, such as GitHub repositories and HTTPS-based data transfers, introduce operational overhead and potential points of failure that must be carefully managed. Currently, the system supports read-only access through FreshService, which limits users to viewing and retrieving information; implementing advanced editing or collaborative capabilities would require substantial additional development. Pipeline monitoring is critical to ensure the reliability of data ingestion, as failures or delays in the pipeline can disrupt system functionality. Furthermore, the system does not currently include automated validation or error-correction mechanisms for incoming data, meaning that administrative oversight remains essential to maintain accuracy. Lastly, the design emphasizes stability and security over real-time responsiveness, which constrains its utility for immediate operational decision-making or dynamic updates.

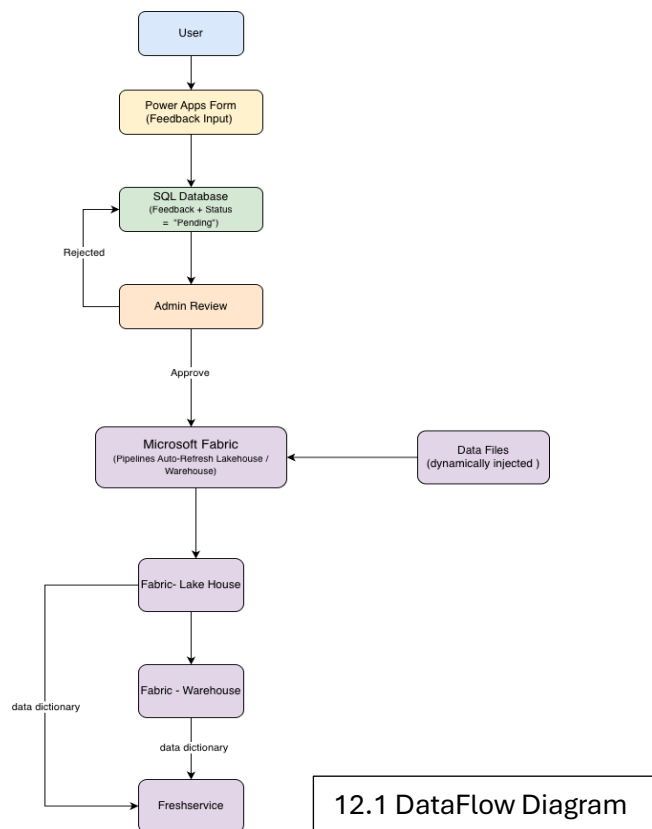
10. Practical Applicability

While the system was initially designed for the MSA ERP environment, its underlying architecture is highly adaptable and can be applied to a wide range of enterprise systems that require structured and governed metadata management. Organizations managing large datasets with multiple stakeholders, approval workflows, and compliance or governance requirements can leverage this model with minimal customization. Beyond ERP platforms, this approach can support data warehouses, business intelligence platforms, and knowledge management systems, providing a centralized framework for metadata tracking, auditing, and reporting. Additionally, the modular design allows organizations to selectively implement features based on their specific needs, whether for operational oversight, data quality assurance, or compliance monitoring. The system's emphasis on standardized processes and clear role definitions ensures that it can enhance collaboration, reduce errors, and improve overall transparency across various organizational contexts.

11. Future Work

Several enhancements are planned to expand the system's functionality and usability. Integration with enterprise metadata catalogs would allow for a more seamless connection with existing organizational data ecosystems, enabling users to access and update metadata directly within familiar platforms. Automated notifications for pending reviews or approvals could improve workflow efficiency and reduce delays in governance processes. AI-assisted quality checks represent another promising area, providing intelligent suggestions or alerts for inconsistencies, missing information, or potential compliance violations. The scope of metadata coverage could also be broadened to include key performance indicators (KPIs), business rules, and additional descriptive attributes, offering a more comprehensive view of organizational data. Finally, as platform capabilities evolve, direct Fabric connectors may replace intermediate storage layers, streamlining data integration and reducing system latency, while also enabling real-time updates and improved scalability. These future developments aim to enhance both the operational effectiveness and strategic value of the system across enterprise environments.

12. Final workflow



12.1 DataFlow Diagram

13. Conclusion

This project presents a fully automated and governance-aligned Data Dictionary system designed to address the challenges of fragmented and manually maintained metadata in the MSA ERP environment. By integrating structured user submissions, administrative review, secure file transport, automated ingestion pipelines, curated warehouse processing, and enterprise-level publication, the system provides a centralized, reliable, and continuously updated framework for metadata management.

The platform resolves a critical organizational problem: inconsistency and ambiguity in interpreting complex ERP datasets. Through version-controlled storage, automated export to GitHub, and integration with Microsoft Fabric, the system ensures that metadata remains accurate, traceable, and accessible to both technical and non-technical users. By streamlining metadata submission, validation, and publication, it enhances documentation quality, supports data governance, and reduces operational inefficiencies associated with ad hoc documentation practices.

While the system is primarily focused on structured metadata management, it establishes a scalable foundation for future enhancements. Potential improvements include AI-assisted quality checks, expanded metadata coverage such as KPIs and business rules, and direct integration with enterprise

metadata catalogs. These developments could further improve operational efficiency, enable more proactive governance, and support advanced analytical workflows.

Overall, the project demonstrates how a structured, automated, and governed approach to metadata management can transform enterprise data practices. It provides a practical, extensible, and sustainable solution that not only meets current organizational needs but also lays the groundwork for long-term improvements in ERP data governance, reporting accuracy, and informed decision-making across the enterprise.

14. References

- [1] Microsoft, *Microsoft Fabric Documentation*, 2025. [Online]. Available: <https://learn.microsoft.com/en-us/fabric/>
- [2] DAMA International, *DAMA-DMBOK: Data Management Body of Knowledge*, 2nd ed., 2017.
- [3] R. Kimball and M. Ross, *The Data Warehouse Toolkit*, Wiley, 2013.
- [4] Microsoft, *Data Governance and Metadata Management Best Practices*, 2025. [Online]. Available: <https://www.microsoft.com/en-us/microsoft-fabric/resources/data-101/what-is-data-governance>
- [5] Microsoft, *Microsoft Power Apps Documentation*, 2025. [Online]. Available: https://learn.microsoft.com/en-us/power-apps/_Microsoft Learn
- [6] Microsoft, *SQL Server Documentation*, 2025. [Online]. Available: https://learn.microsoft.com/en-us/sql/?view=sql-server-ver17_Microsoft Learn
- [7] Microsoft, *What is an On-Premises Data Gateway?*, Microsoft Learn, 2025. [Online]. Available: https://learn.microsoft.com/en-us/data-integration/gateway/service-gateway-onprem_Microsoft Learn
- [8] Microsoft, *Manage an On-Premises Data Gateway in Power Apps*, Microsoft Learn, 2022. [Online]. Available: <https://learn.microsoft.com/en-us/power-apps/maker/canvas-apps/gateway-management>